

MAMMALS

No class of vertebrates has conquered the world's wild places like mammals. Warm blood, a big brain, and well-nurtured offspring help them succeed practically wherever life is possible. They thrive in places too cold for reptiles, and a fifth of their kind—the bats—can fly as expertly as birds. Others are runners,

climbers, burrowers, and swimmers. Oceangoing mammals, such as the whales, are the biggest animals on the planet, and a few marine mammals can even dive up to a mile below the water's surface, holding their breath for an hour or more.

Anatomy

Mammals evolved from lizardlike ancestors—a transition that required considerable changes to the skeleton. Parts of the reptilian jaw became miniaturized in mammals and reassigned to the middle ear to help transmit sound signals. This left mammals with just a single pair of bones in their lower jaw—but resulted in a stronger, more flexible, chewing action. The reptiles' uniform teeth changed too. Mammals now have ridged, grinding molars at the back and stabbing canines at the front—helping some to eat tough, fibrous plants and others to be bone-crushing predators.

Proficient feeding provides fuel for the body, and a mammal's metabolism generates a lot of heat. This makes mammals warm blooded, while a complex system of self-regulation keeps their temperature constantly high. Most mammals have a coat of hairs to trap this warmth close to the body, and because of this, unlike cold-blooded vertebrates, they can stay active even when their surroundings change. Their high oxygen demand is satisfied by a powerful four-chambered heart and lungs that are inflated by a muscular diaphragm located below the chest cavity.

Reproduction

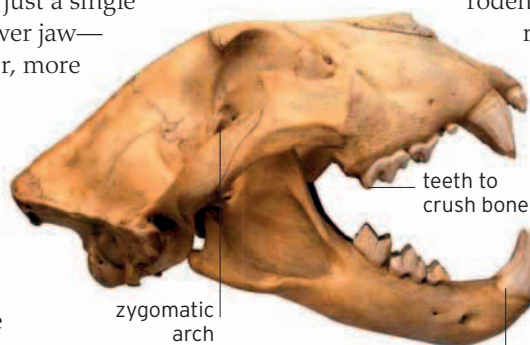
Mammals are typically devoted parents. Most give birth to live young after a pregnancy period, during which the unborn are carried in the womb and nourished by an advanced blood-filled organ called the placenta. Marsupials are born at a more premature stage of their development, and in monotremes, the young hatch from eggs. However, all mammalian mothers suckle their offspring after birth with nutrient-rich milk secreted from the

mammary glands. Many mammals give birth to one or a few offspring and can therefore give a relatively large amount of care to each individual. The single young of a large ape, such as an orangutan, could remain with its mother for many years. In contrast, small rodents—with a speedy reproductive cycle—can produce litters of a dozen babies, which are ready to breed themselves in just a few weeks.

Behavior

Like their early ancestors, most living mammals use well-developed senses of hearing and smell to interact with others as well as to hunt at night. Although some, such as monkeys, evolved to be

active during the day, the sense of smell is still important in the life of most mammals, and many communicate socially by using scent. In a warm-blooded body, the temperature-sensitive



△ LION'S SKULL

Mammals have especially effective jaws—and specialized teeth for crunching or stabbing. The strong zygomatic arch of a lion's skull supports muscles for a powerful bite.

canine teeth for gripping or tearing



△ PIGGYBACKING LITTER

Litter size varies in mammals. Opossums have up to a dozen infants—carried first, like most marsupials, in their mother's pouch, but then on her back.

brain has evolved to carry out sophisticated tasks, making mammalian behavior highly complex. This means that mammals can learn as well as solve problems. Infants pick up life skills by watching their parents or other individuals, but mammals vary in their social behavior. Most cats, for example, are solitary, fiercely territorial, and only form temporary bonds for the purpose of mating. Conversely, many hooved mammals socialize in herds and, in places, gather to form some of the biggest aggregations of wildlife seen on the planet.



◁ FIERCE BATTLE

Mammalian social behavior varies with species and sex. Male gemsboks—a type of desert antelope—clash in disputes over mates, while females socialize in small herds.